DATA ANALYST CAPSTONE PROJECT

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OUTLINE



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- Methodology
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- Discussion
  - Findings & Implications
- Conclusion
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EXECUTIVE SUMMARY



- JavaScript, SQL, HTML/CSS, and TypeScript lead current use and future interest
- Technology preferences highlight key training priorities:
  - PostgreSQL leads database use and future interest
  - React and Node.js remain leading web technologies
  - Go, Rust, and Redis show growing interest
- Hybrid and remote work are more popular than fully in-person work
- The USA has the highest compensation in the analyzed data. Washington, DC leads US job postings
- Swift users earn the highest average salary among the 10 languages compared, at over $120k. Pay generally rises with experience

INTRODUCTION



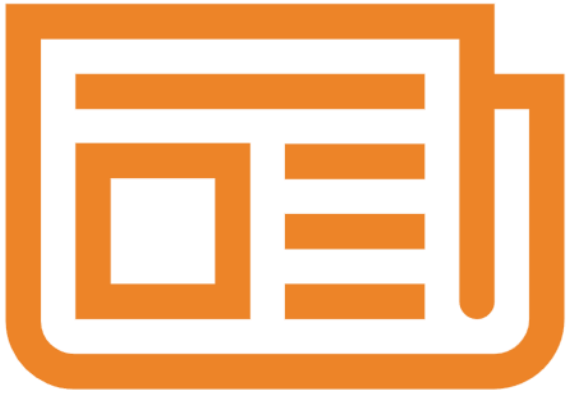
- Purpose: identify trends for in-demand skills
- Target Audience: internal executive leadership, human resources team, learning & development team
- Value of analysis:
  - Help to align market demands
  - Proactive talent planning
  - Data-driven decision-making

QUESTIONS INCLUDE:



- Which programming languages are most in demand?
- Which database technologies are currently most sought after?
- Which Integrated Development Environments (IDEs) are the most popular?

METHODOLOGY



Data Sources

- **2024 Stack Overflow Developer Survey**
- Relevant websites and API-accessible datasets

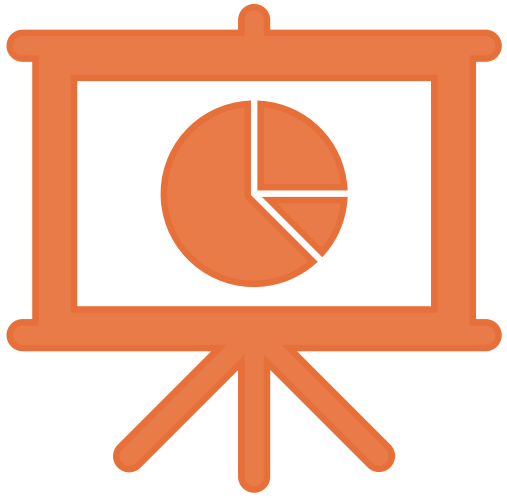
Data Collection

- Web scraping and API requests
- Data imports from CSV files, Excel spreadsheets, and databases

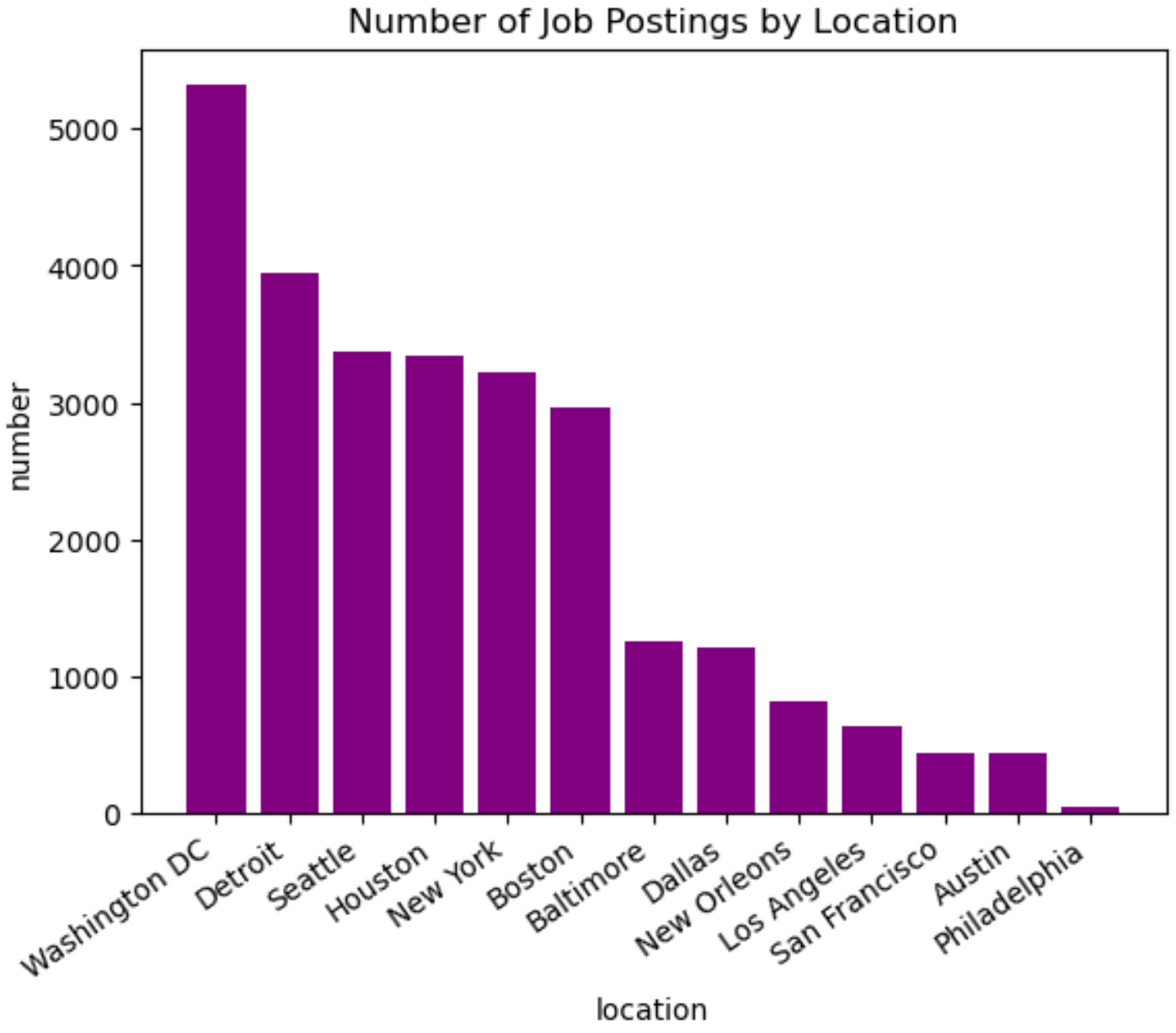
Data Wrangling

- Removed duplicates and identified missing values
- Filled missing numerical values using the mean or median, or dropped records where appropriate
- Standardized country names and shortened lengthy labels
- Simplified education-level labels for clearer visualizations
- Split multi-value language and platform fields into separate columns

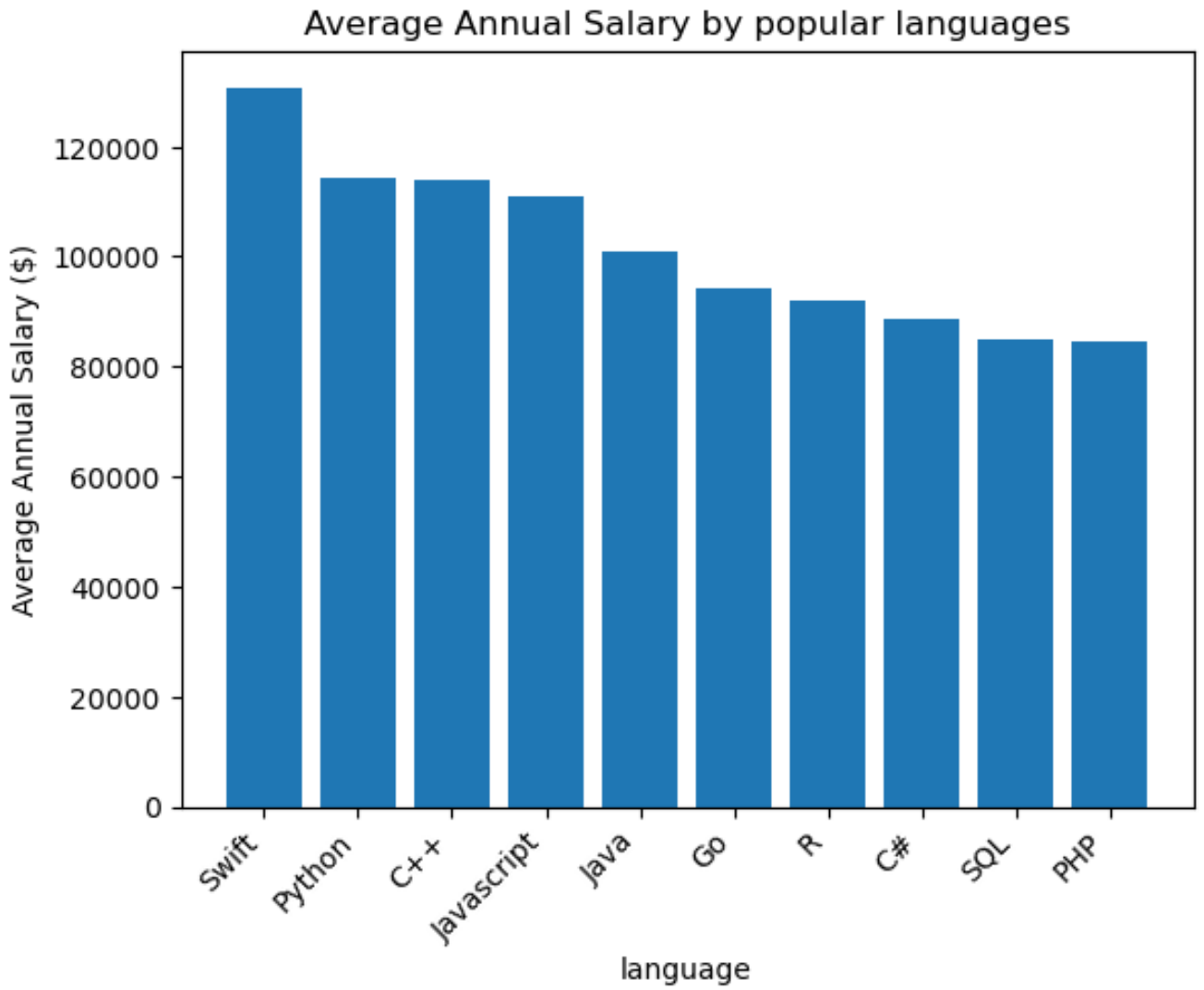
RESULTS



JOB POSTINGS

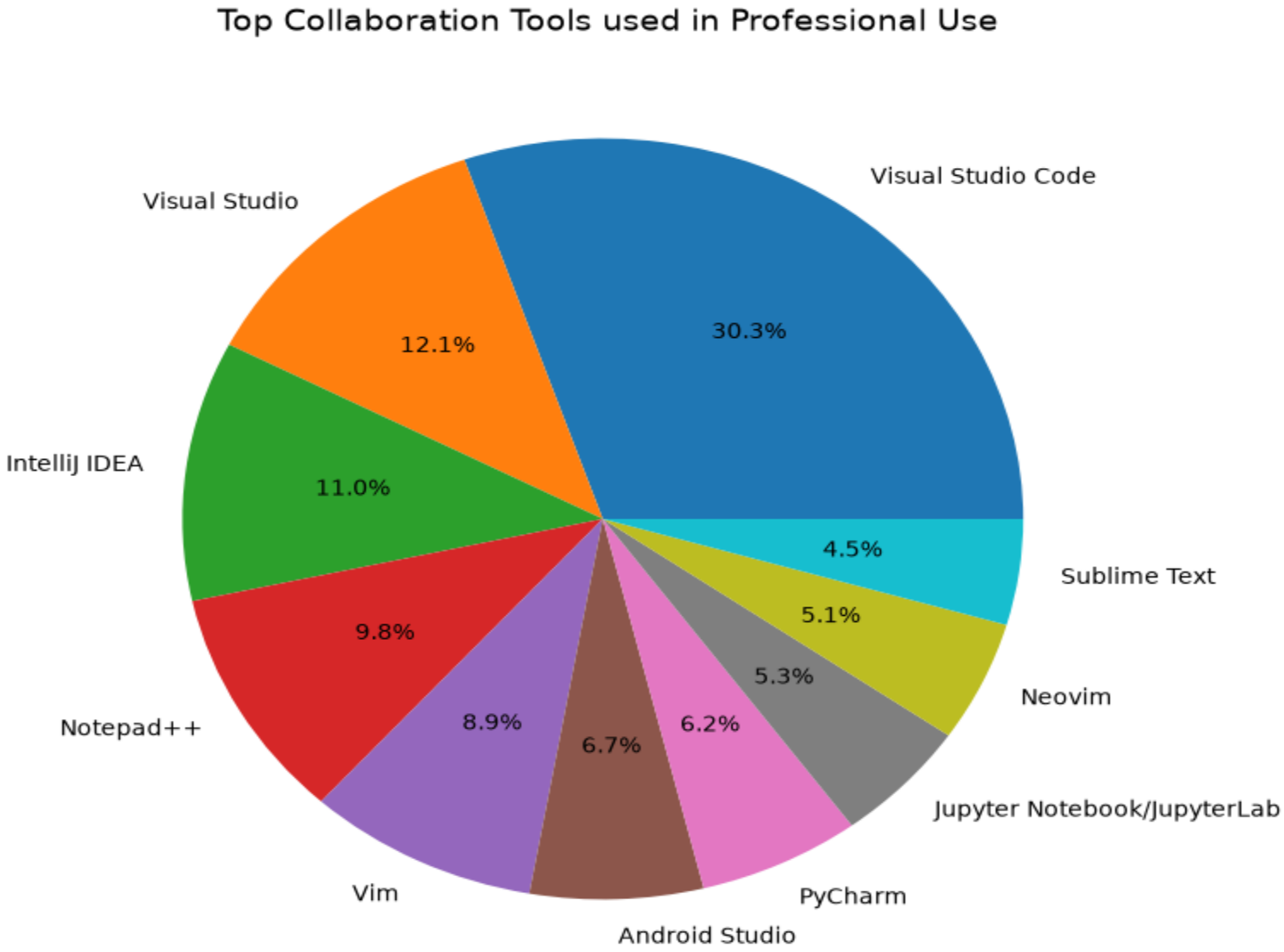


POPULAR LANGUAGES



Data Source: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/datasets/Programming_Languages.html

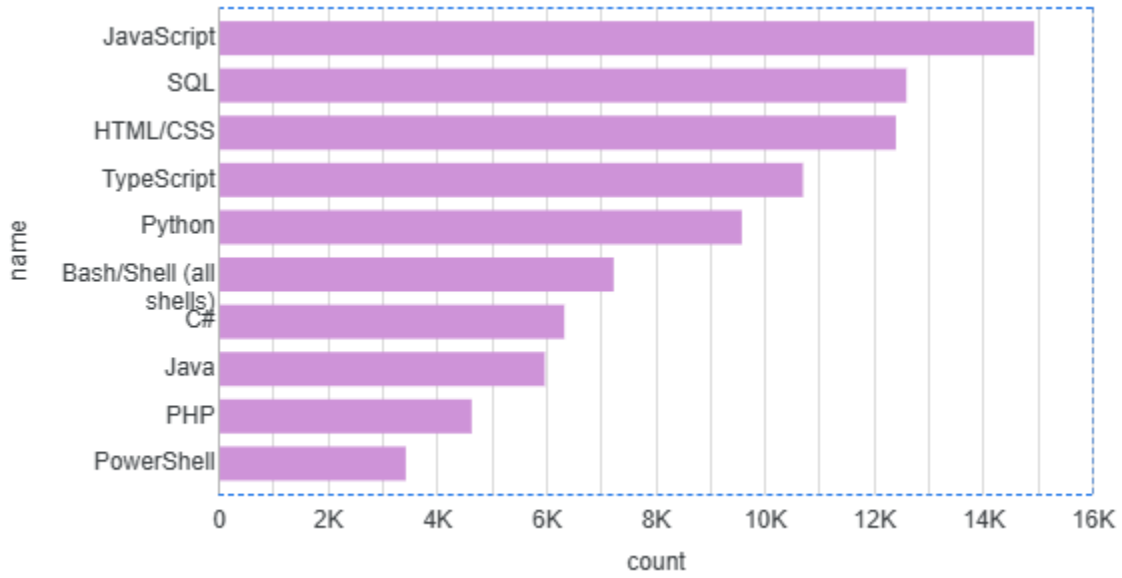
POPULAR COLLABORATION TOOLS



PROGRAMMING LANGUAGE TRENDS

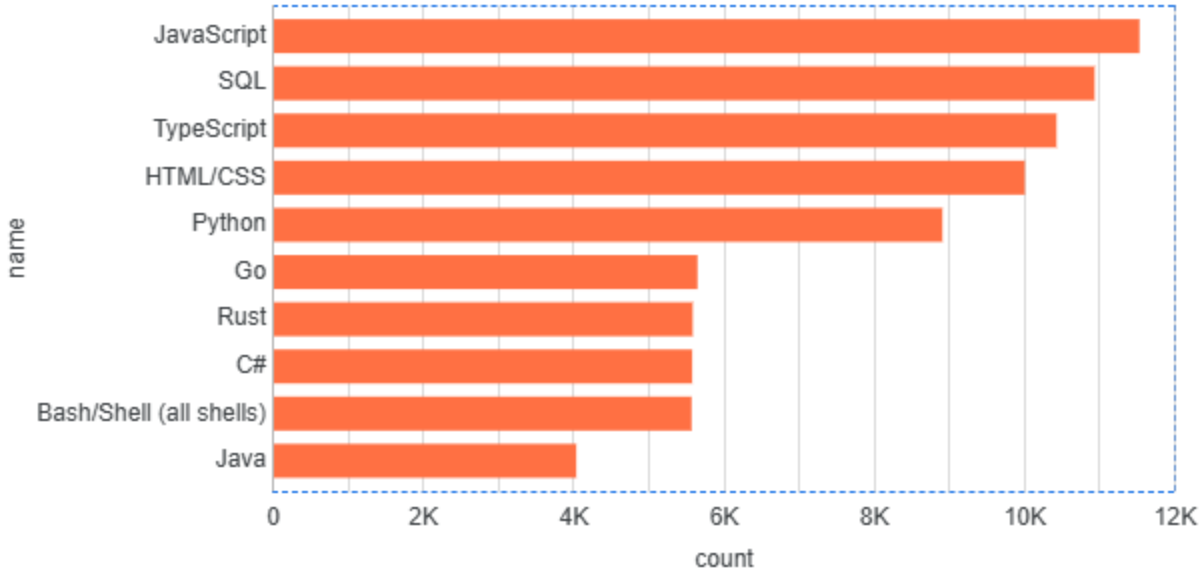
Current Year

- Top 10 Programming Language used



Next Year

- Top 10 Programming Language Desired



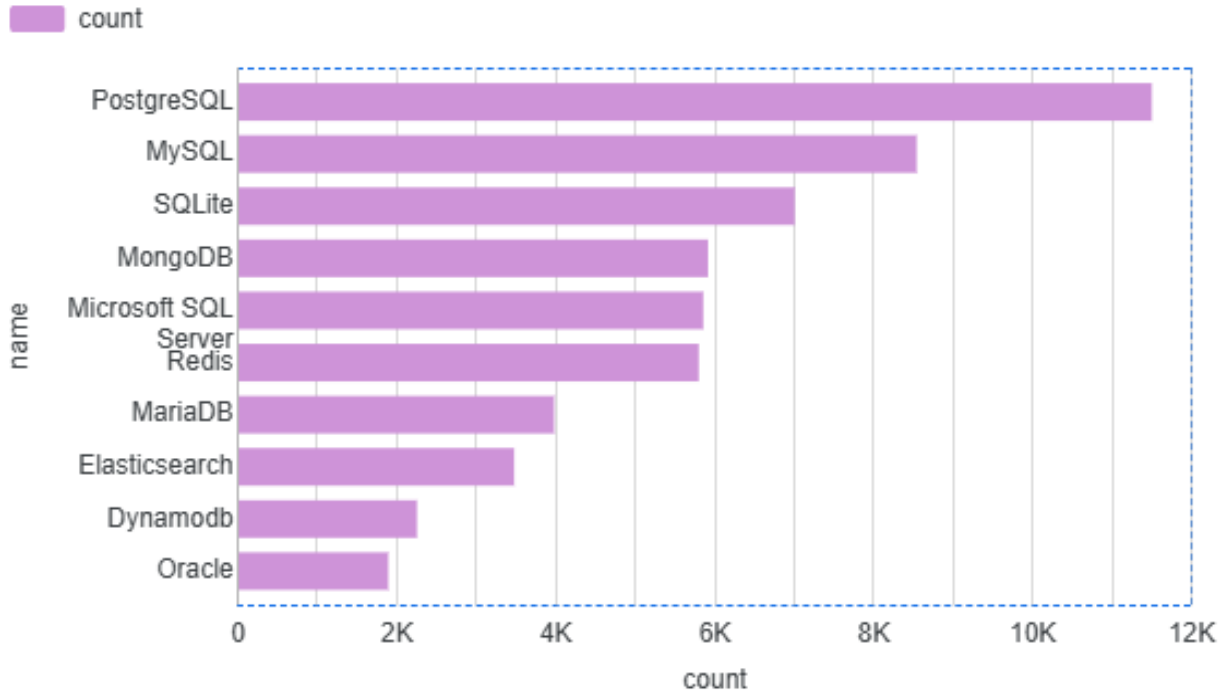
PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

	Findings	Implications
1	**JavaScript, SQL, HTML/CSS, and TypeScript** remain the top 4 languages in both current use and future interest	Keep investing heavily in JavaScript, TypeScript, and SQL training because client demand will stay strong
2	**Python** remains stable demand	Continue Python projects and training
3	**Go and Rust** gained rising interest	Start building teams and pilot projects around Go and Rust to meet future high-performance needs
4	**Bash/Shell** drops from 7,244 to 5,582, **Java** falls from 5,982 to 4,048	Reduce new hiring and training for Java, PHP, and PowerShell, focusing instead on supporting older systems while clients transition away
5	**PHP along with PowerShell** completely drop out of the top 10 desired languages	

DATABASE TRENDS

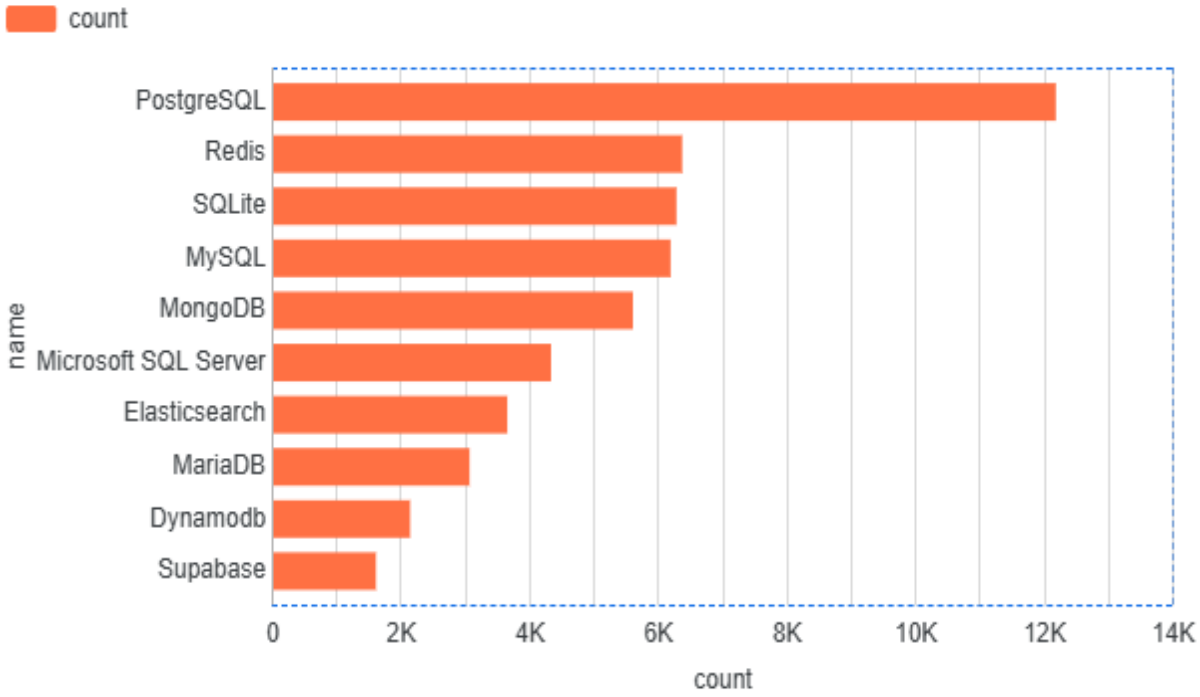
Current Year

- Top 10 Databases Used



Next Year

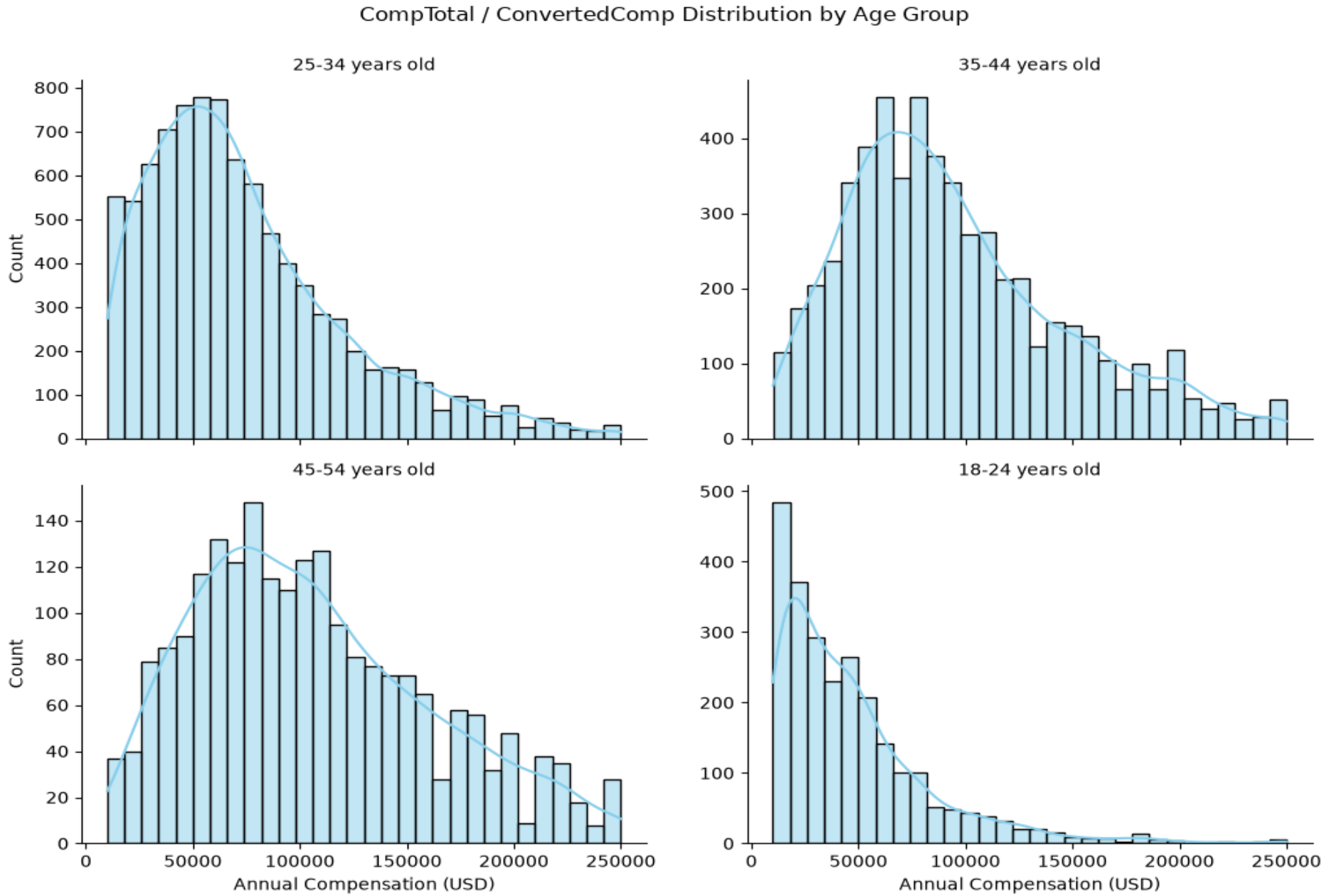
- Top 10 Desired Databases



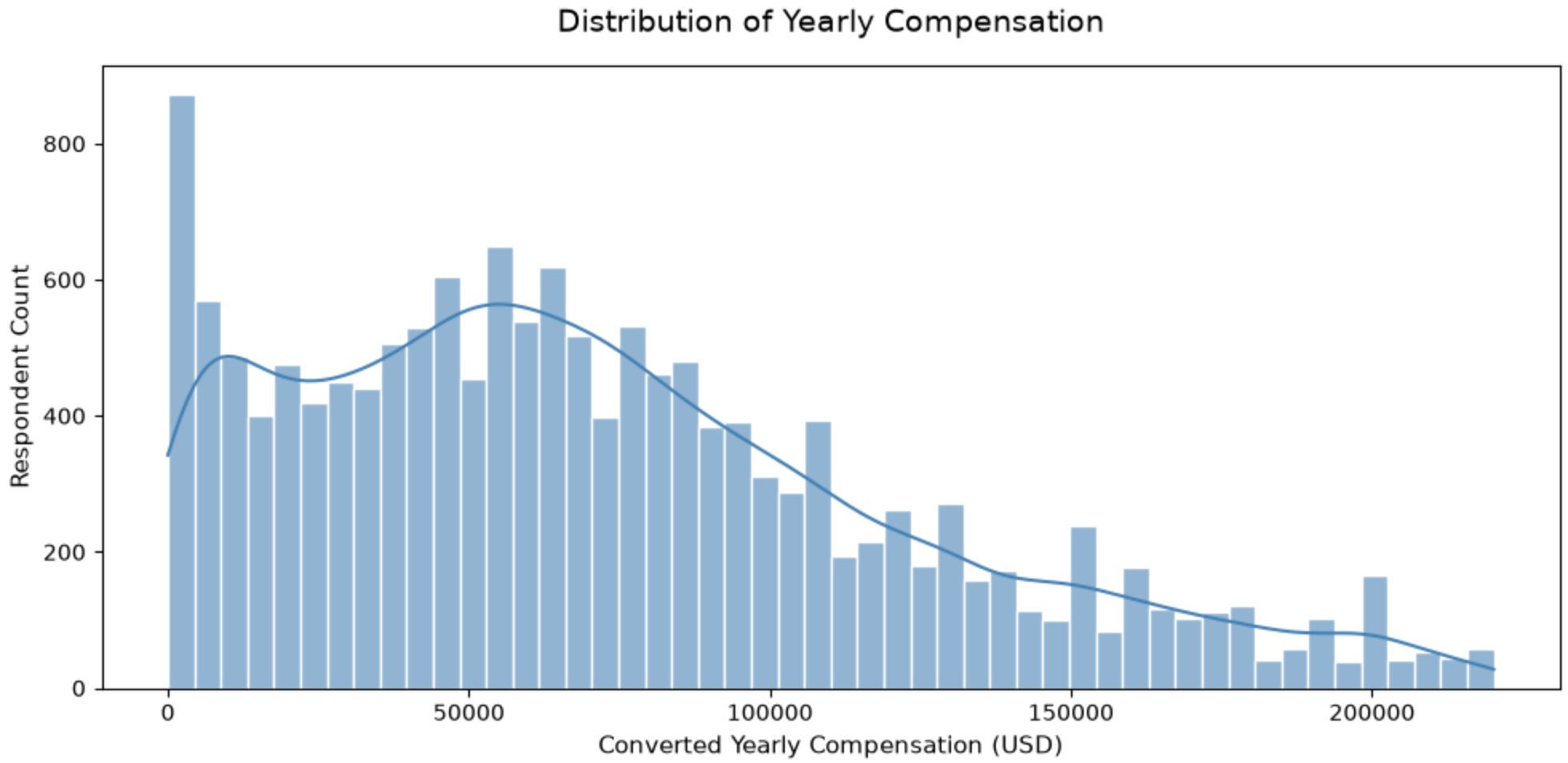
DATABASE TRENDS - FINDINGS & IMPLICATIONS

	Findings	Implications
1	**PostgreSQL** is the most used database and grows even more in popularity next year	Teach developers PostgreSQL and make it the main choice for new client projects
2	**Redis** moves from 6th place to 2nd place on the wish list	Help teams learn Redis to build faster apps and speed up data caching
3	Fewer developers want to use **MySQL and SQLite** in the future	Start shifting older MySQL projects toward modern PostgreSQL systems
4	**Supabase** enters the top 10 list, while **Oracle** drops out completely	Spend less time on Oracle databases because client interest is fading away

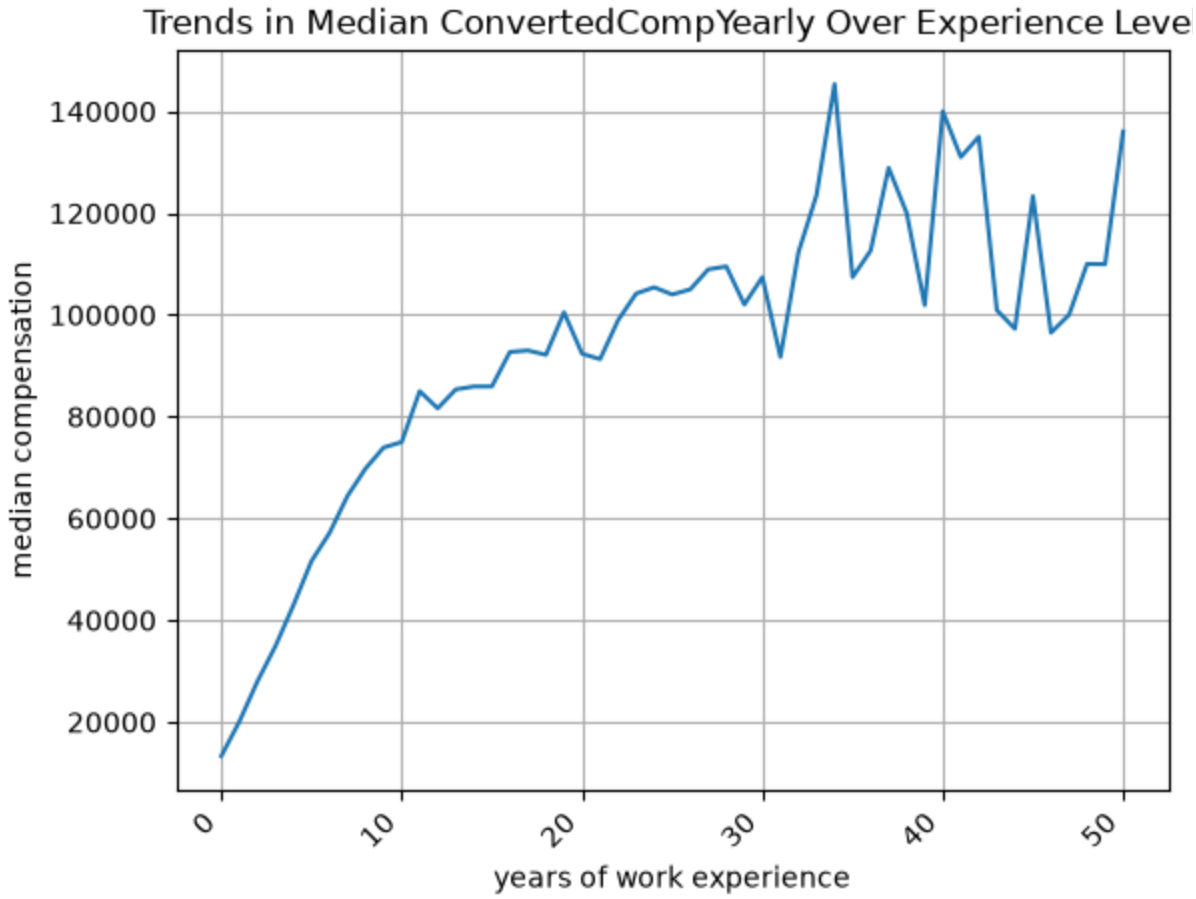
ANNUAL COMPENSATION (USD) TRENDS

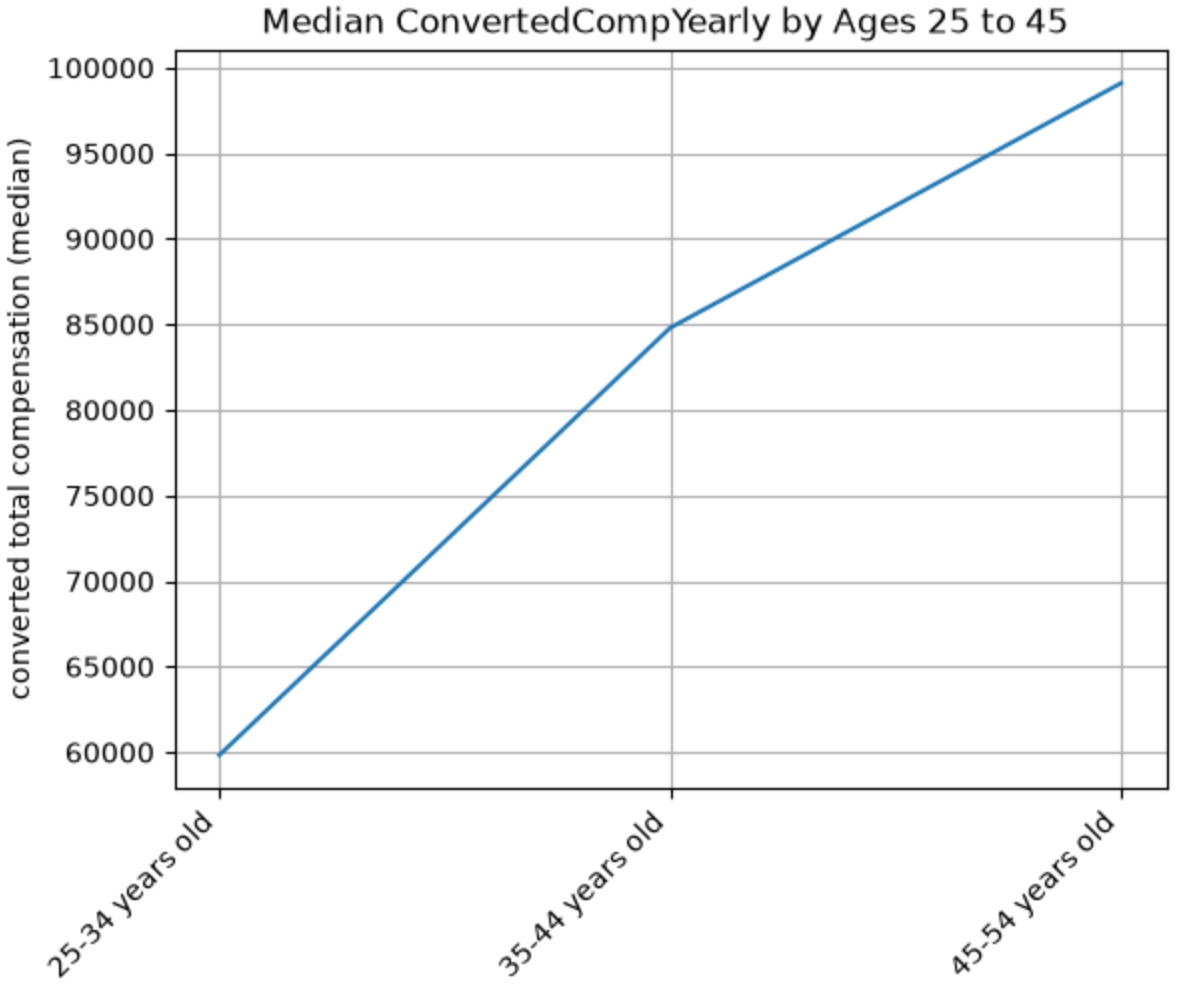


ANNUAL COMPENSATION (USD) TRENDS (CONTINUED)



ANNUAL COMPENSATION (USD) TRENDS (CONTINUED)

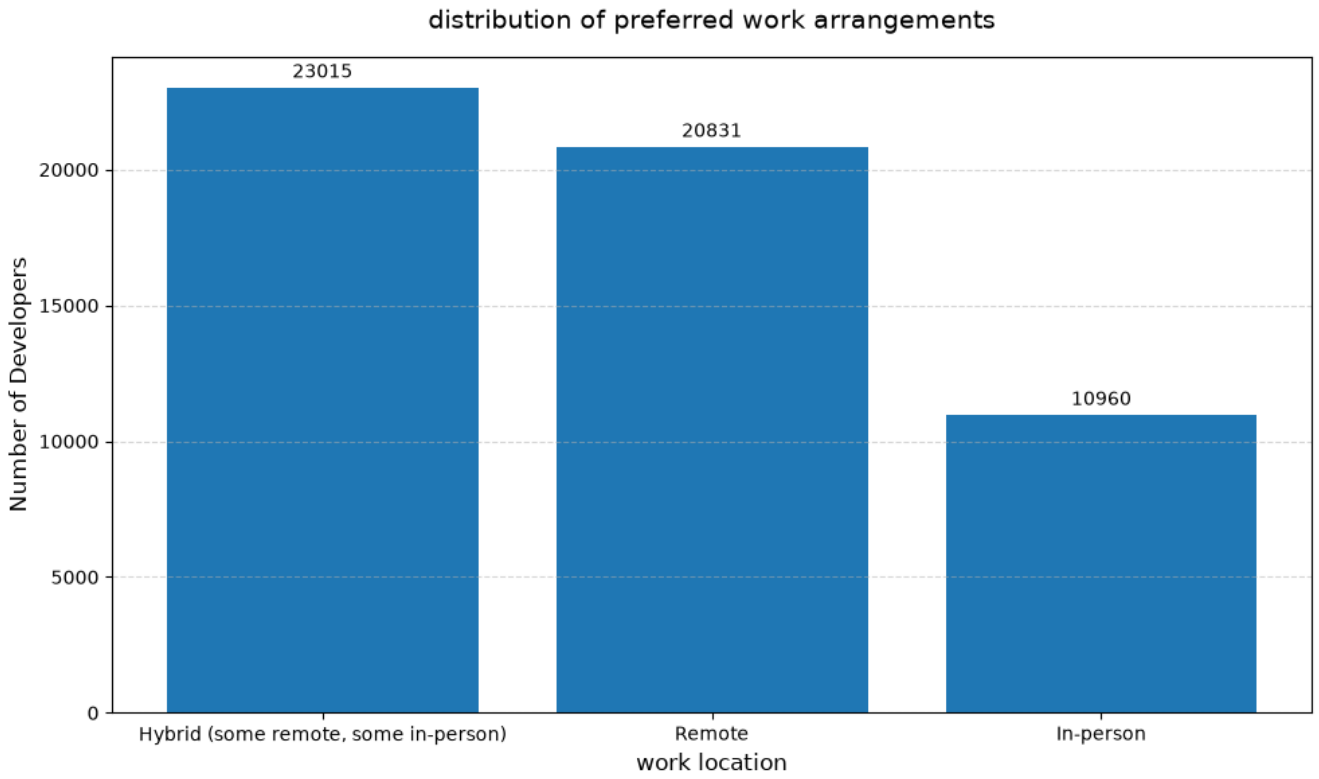




ANNUAL COMPENSATION (USD) TRENDS (CONTINUED)

- Median salary grows steadily from about $60k at ages 25–34 to nearly $100k at ages 45–54
- Pay increases fastest during the first 10 to 15 years of work, rising from under $20k to over $80k
- After around 20 years of experience, median pay levels off near $100k and fluctuates more due to smaller sample sizes
- Younger groups (18–24 and 25–34) cluster tightly at lower salaries, while older groups (45–54) spread out much wider into higher pay brackets ($100k to $200k+)

JOB RELATED TRENDS

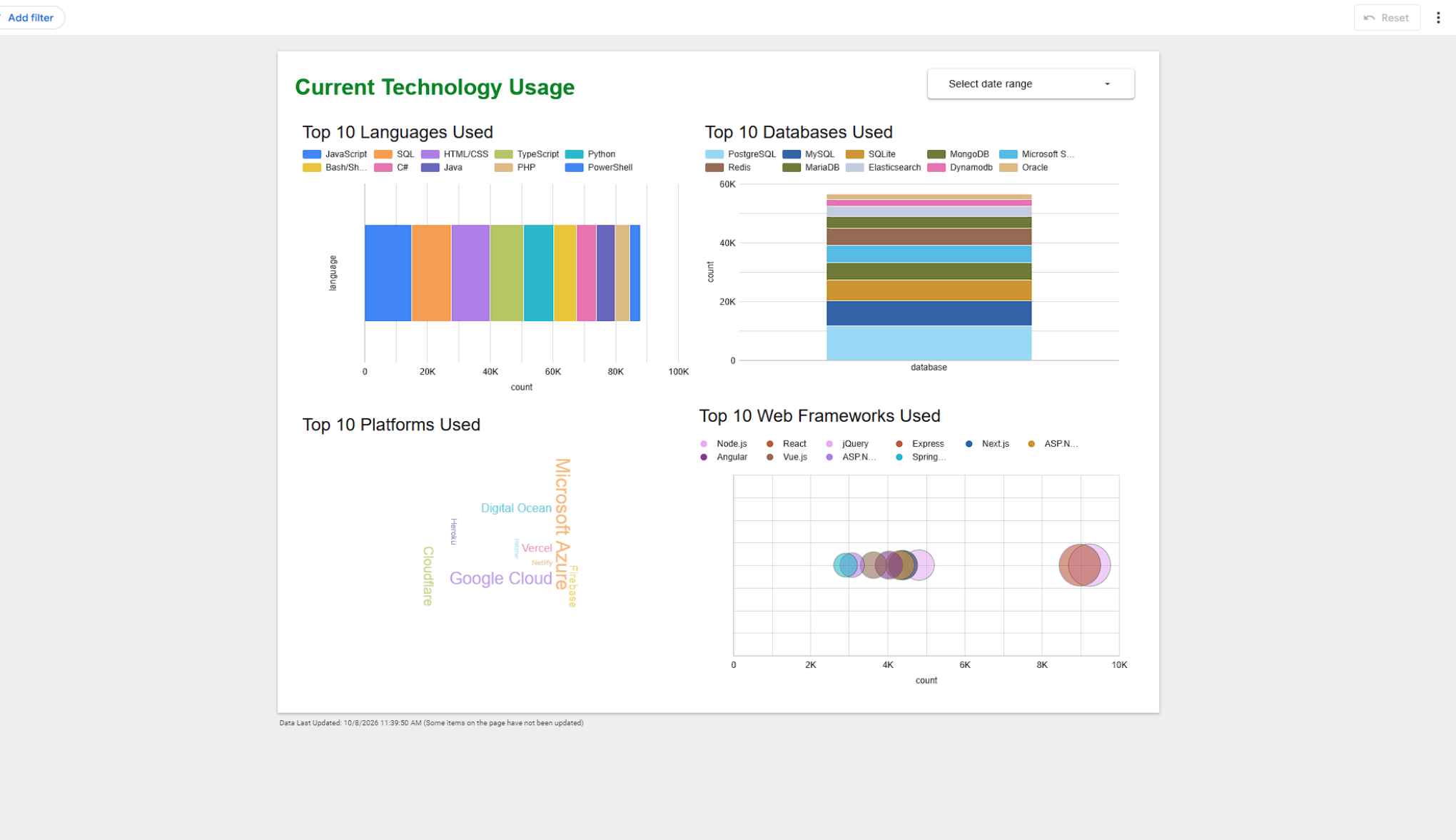


- **Hybrid & Remote** is the most popular
- People have less interest for in-person roles in IT fields

DASHBOARD



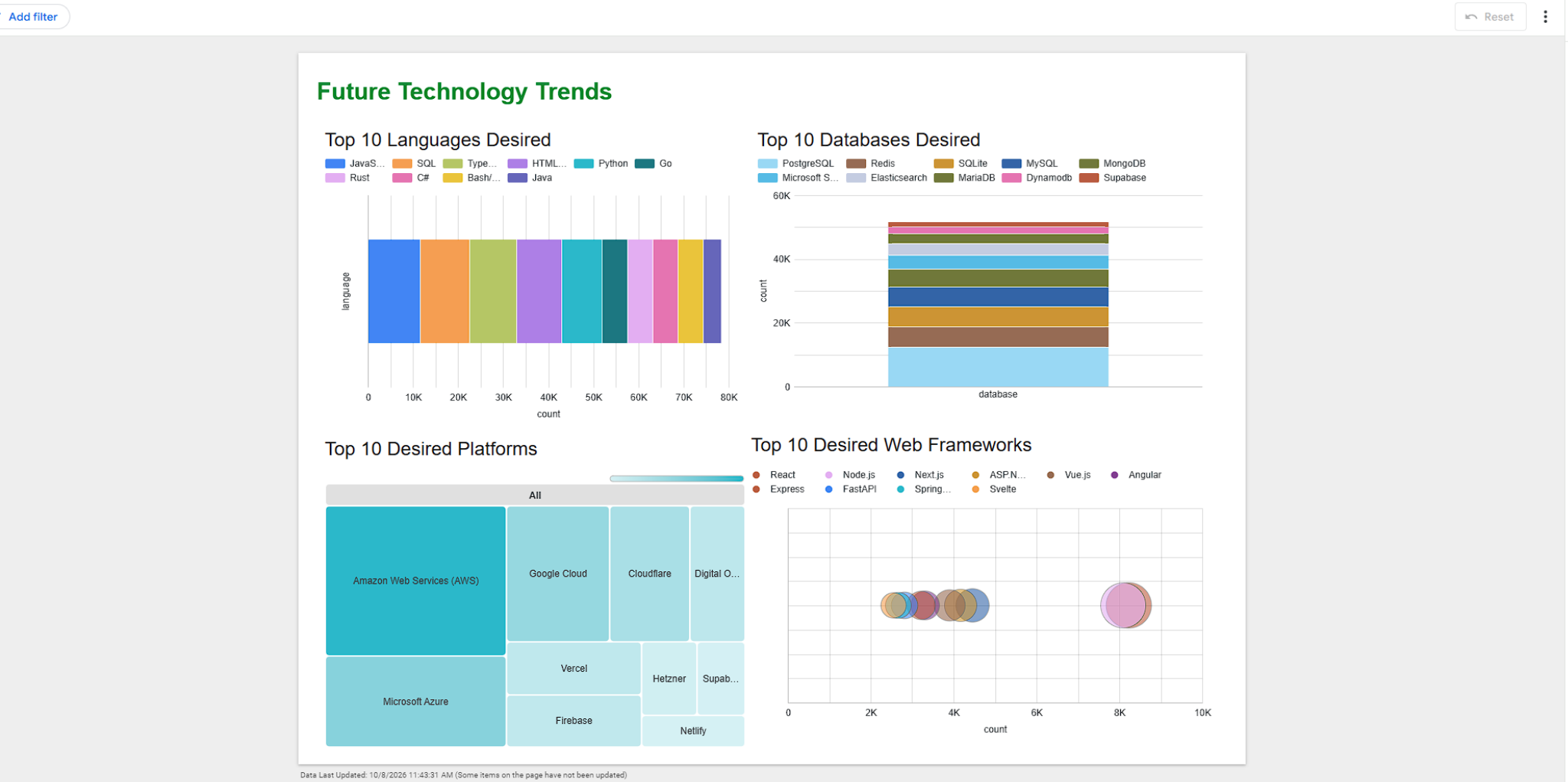
DASHBOARD TAB 1: CURRENT TECHNOLOGY USAGE



CURRENT TECHNOLOGY TRENDS (CONTINUED)

- JavaScript and SQL represent the dominant active languages, forming the backbone of current front-end and relational data operations
- Node.js and React lead production environments, establishing a JavaScript-centered full-stack standard for current client projects
- PostgreSQL is the most widely adopted database, serving as the industry benchmark for structured data workloads.
- Microsoft Azure and Google Cloud are the top platforms in active use, reflecting strong enterprise multi-cloud and infrastructure deployments

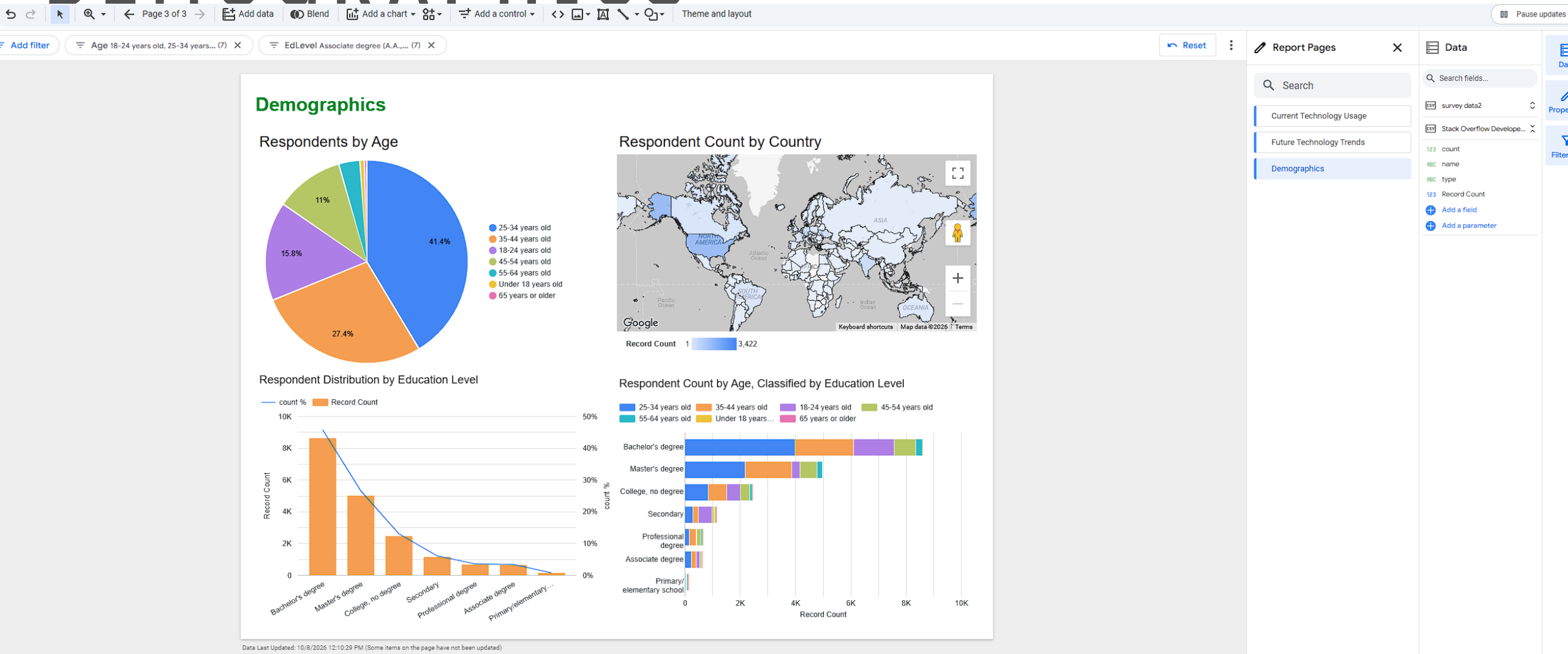
DASHBOARD TAB 2: FUTURE TECHNOLOGY TRENDS



FUTURE TECHNOLOGY TRENDS (CONTINUED)

- **JavaScript, SQL, TypeScript, and HTML/CSS** dominate developer demand, highlighting the dual need for front-end versatility and structured data querying
- **React and Node.js** remain the top web frameworks, solidifying a unified JavaScript/TypeScript full-stack ecosystem
- **PostgreSQL** leads as the preferred database, driven by strong enterprise adoption of modern, open-source relational systems
- **AWS** holds the #1 spot among cloud platforms, remaining the primary deployment foundation for enterprise solutions

DASHBOARD TAB 3: DEMOGRAPHICS



DEMOGRAPHICS (CONTINUED)

- Most respondents are **25 to 34 years old** (41.3%). The second largest group is 35 to 44 years old (27.3%). Very few respondents are 65 or older
- Most respondents live in the **United States**
- A **Bachelor's degree** is the most common education level (around 46%). A **Master's degree** is the second most common level. Doctoral and professional degrees are rare (3.5%)
- **People aged 25 to 34 with a Bachelor's degree** make up the biggest single group

DISCUSSION



- Technology: JavaScript and SQL lead, while Go, Rust, and Redis gain interest
- Adoption: PostgreSQL leads current use and future interest. Older systems still need support
- Salaries: Pay varies by experience, country, and language
- Work preferences: Developers favor hybrid and remote roles
- Limitations: Survey interest does not guarantee future adoption or hiring demand

OVERALL FINDINGS & IMPLICATIONS

Findings

- **JavaScript, SQL, HTML/CSS, and TypeScript** lead both current use and future interest. **React and Node.js** remain leading web technologies
- **PostgreSQL** leads database use and future interest. **Redis, Go, and Rust** show growing interest
- **Visual Studio Code** is the most popular development tool, at 30.3%
- **Hybrid and remote work** are more popular than fully in-person work
- The USA has the highest compensation in the analyzed data. Washington, DC has the most US job postings
- Swift users have the highest average salary among the 10 languages compared, at over $120k. Median pay rises with experience, then levels off around 20 years

OVERALL FINDINGS & IMPLICATIONS

Implications

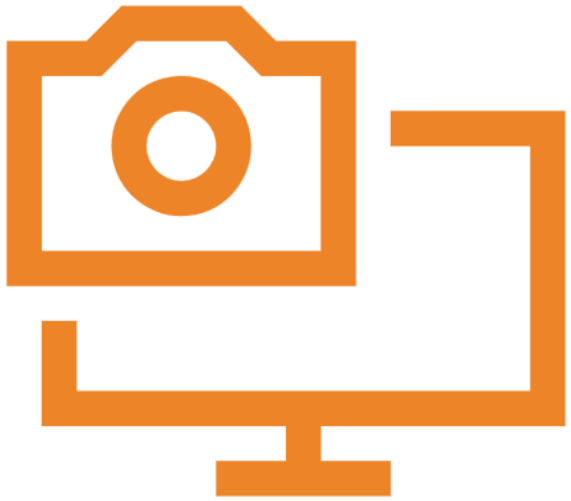
- Prioritize JavaScript, TypeScript, SQL, Python, and PostgreSQL training
- Explore Go, Rust, and Redis through small pilot projects
- Offer hybrid and remote options to attract developers
- Use salary findings as benchmarks, not guarantees. Consider location and experience when comparing pay
- Treat lower future interest as a signal to review priorities, not a reason to immediately replace existing systems

CONCLUSION



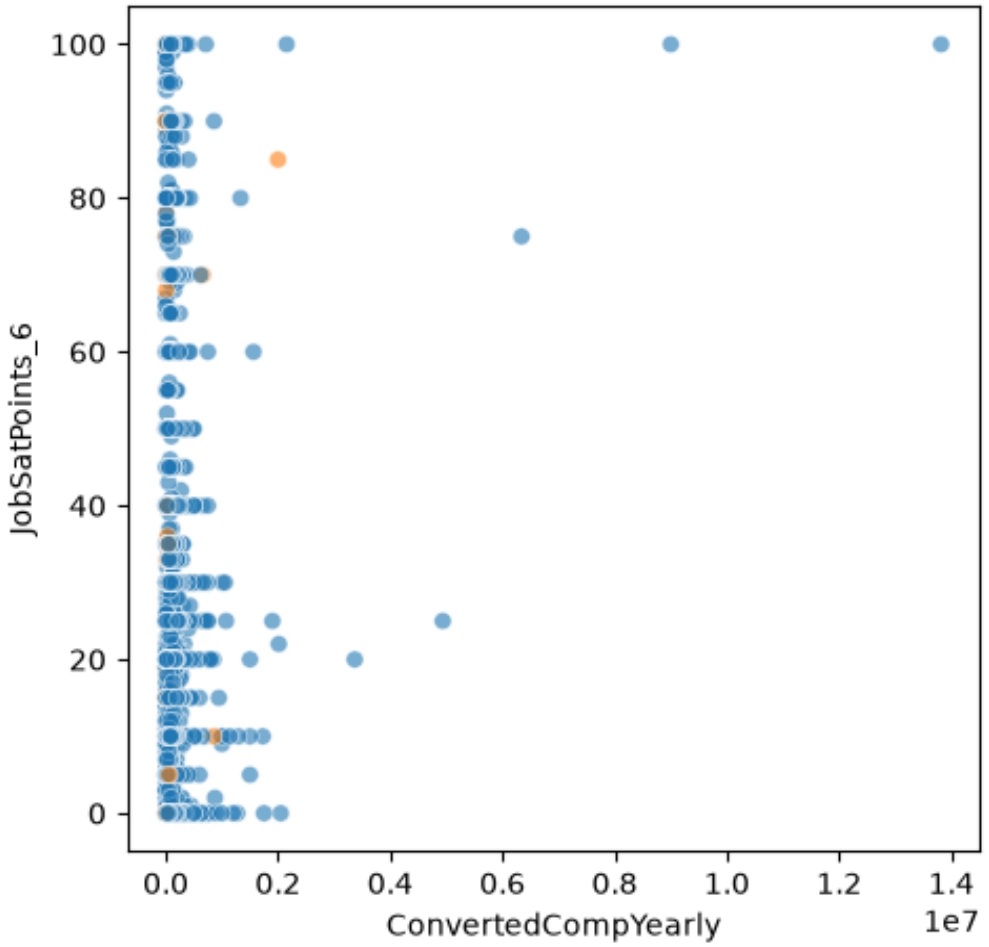
- JavaScript, SQL, and PostgreSQL remain key priorities for developer training
- Growing interest in Go, Rust, and Redis highlights opportunities to build new skills
- Hybrid and remote work are important considerations for hiring
- Salary differences show why location and experience matter when comparing pay
- Use these findings to guide training, hiring, and technology choices, while considering business needs and existing systems

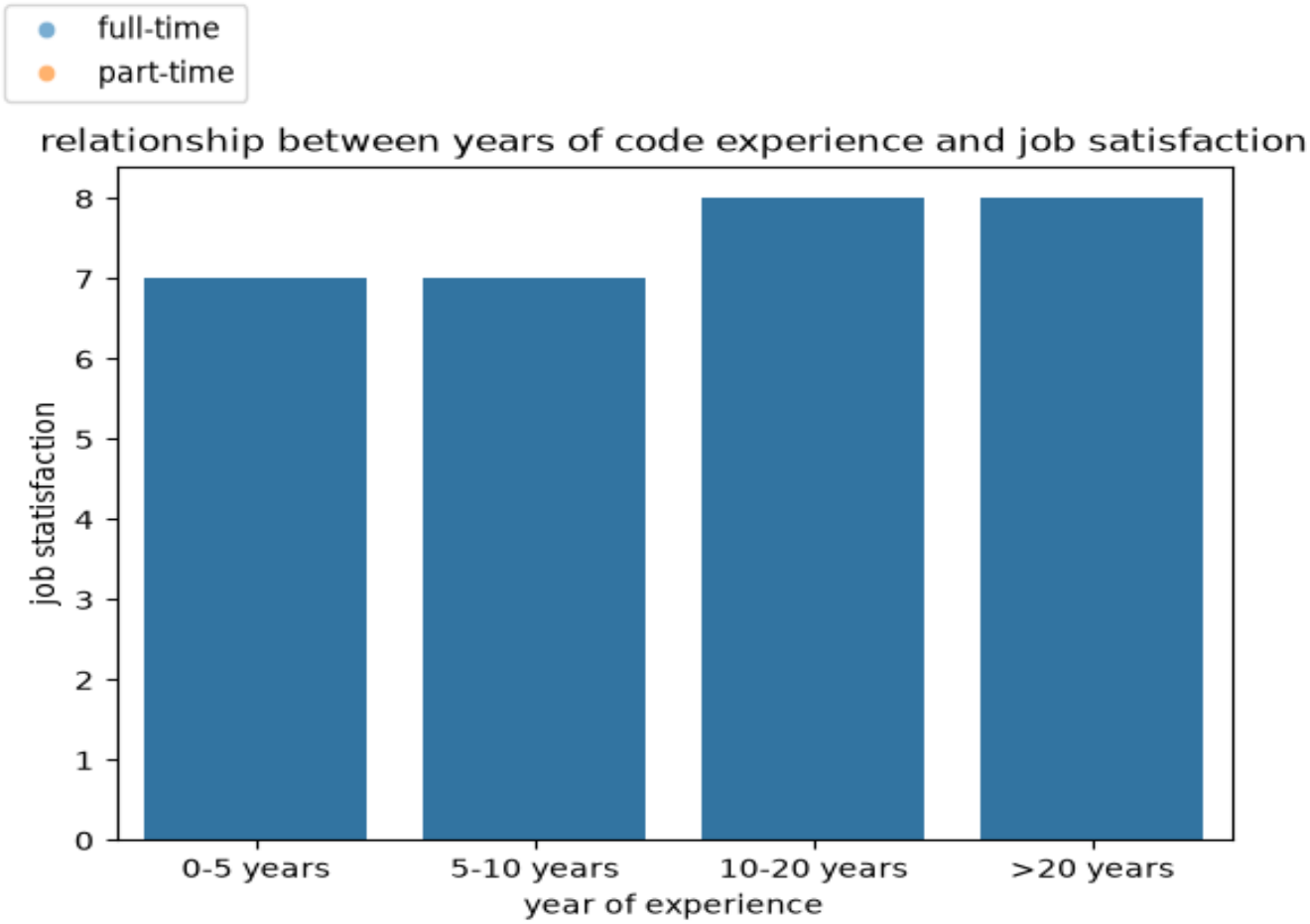
APPENDIX



APPENDIX

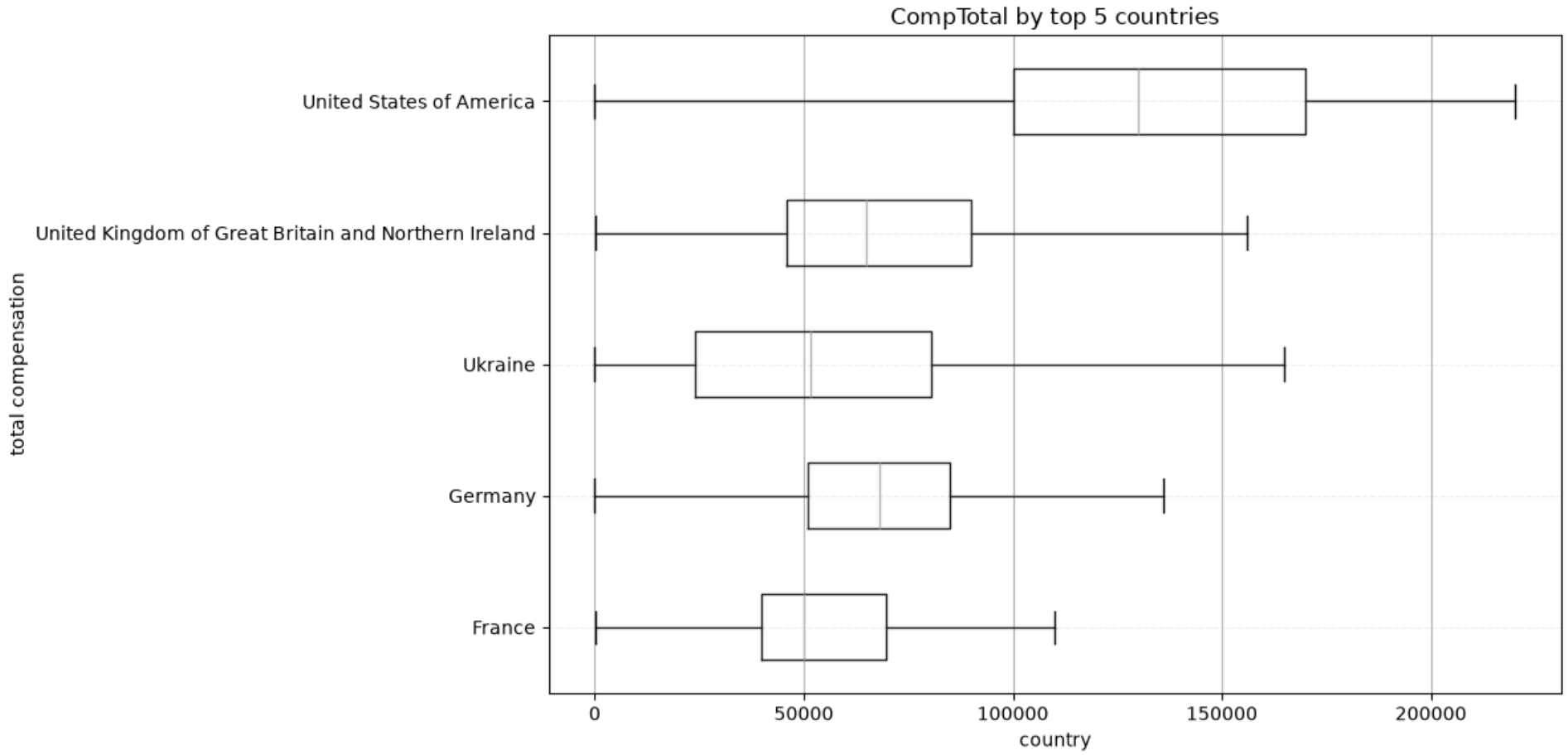
Compensation vs. Job Satisfaction by Employment Type



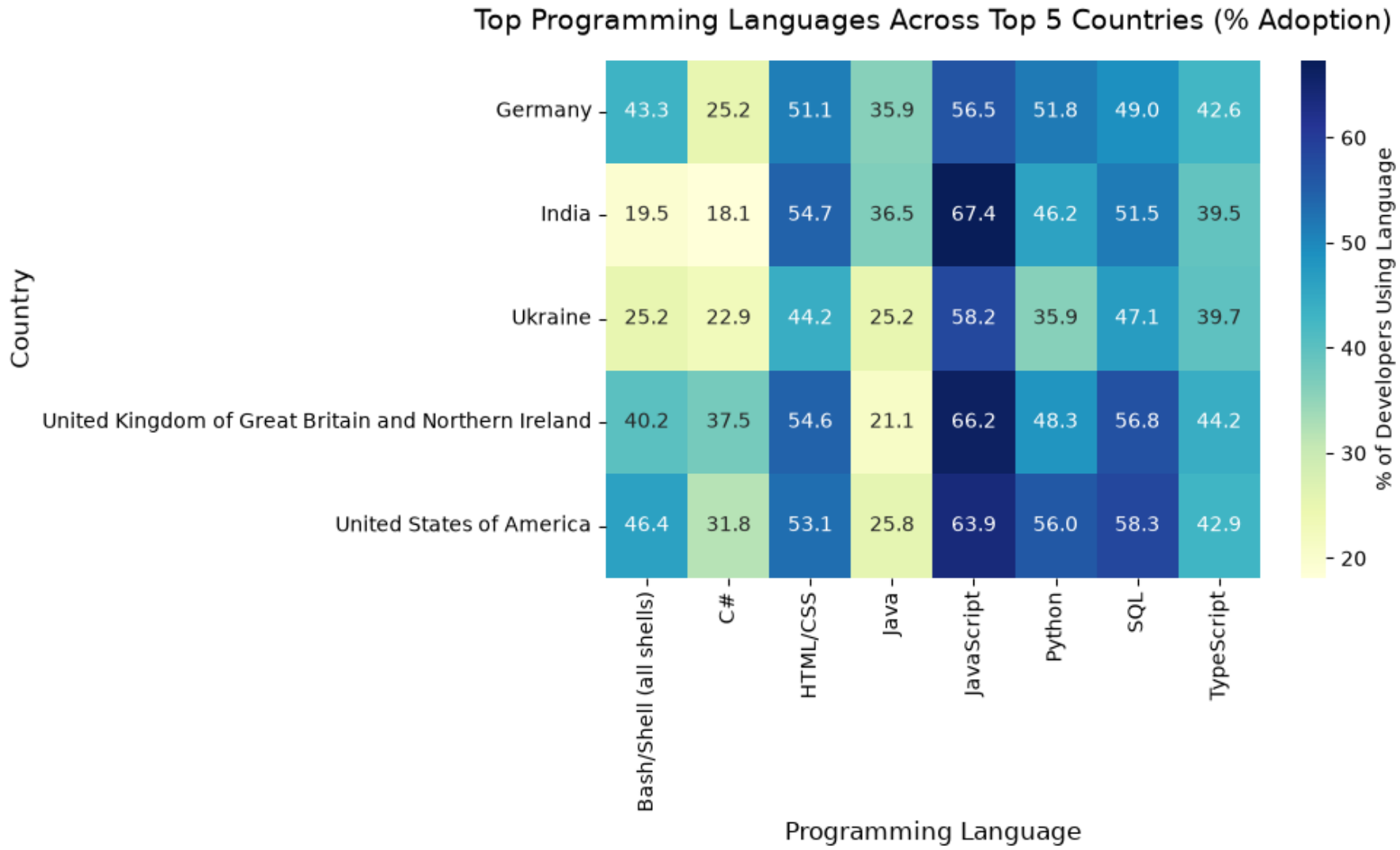


APPENDIX(CONTINUED)

Total Compensation by Top 5 Countries



APPENDIX(CONTINUED)



APPENDIX(CONTINUED)

